$$2.945 = 2 + \frac{9}{10} + \frac{4}{100} + \frac{5}{1000}$$

$$= 2 + \frac{900}{1000} + \frac{40}{1000} + \frac{5}{1000} \text{ (making denominators common)}$$

$$= 2 + \frac{945}{1000}$$

$$= \frac{2945}{1000} \text{ or } \frac{589}{200} \text{ which is required}$$

(In the above, is it possible to write directly $2.945 = \frac{2945}{1000}$?)

Example 2.4 Convert the following decimal numbers in the form of $\frac{p}{q}$, where p and q are integers and $q \neq 0$: (i) 0.35 (ii) 2.176 (iii) – 0.0028

Solution

(i) $0.35 = \frac{35}{100} = \frac{7}{20}$

(ii) $2.176 = \frac{2176}{1000} = \frac{272}{125}$

(iii) $-0.0028 = \frac{-28}{10000} = \frac{-7}{2500}$

2.3.5 Conversion of Non-terminating and recurring decimals into Rational Numbers

It was very easy to handle a terminating decimal. When we come across a decimal such as 2.4, we get rid of the decimal point, by just divide it by 10.

Thus $2.4 = \frac{24}{10}$, which is simplified as $\frac{12}{5}$. But, when we have a decimal such as $2.\overline{4}$, the problem is that we have infinite number of 4s and hence will need infinite number of 0s in the denominator. For example,

$$2.4 = 2 + \frac{4}{10}$$

$$2.44 = 2 + \frac{4}{10} + \frac{4}{100}$$

$$2.444 = 2 + \frac{4}{10} + \frac{4}{100} + \frac{4}{1000}$$

How tough it is to have infinite 4's and work with them. We need to get rid of the infinite sequence in some way. The good thing about the infinite sequence is that even if we pull away one , two or more 4 out of it, the sequence still remains infinite.

Let $x = 2.\overline{4}$ ...(1)

Then, $10x = 24.\overline{4}$ ...(2) [When you multiply by 10, the decimal moves one place to the right but you still have infinite 4s left over).

Subtract the first equation from the second to get,

$9x = 24.\overline{4} - 2.\overline{4} = 22$ (Infinite 4s subtract out the infinite 4s and the left out is $24 - 2 = 22$)

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